

# Mayatra Vidhi

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**IISER-THIRUVANANTHAPURAM**

**BS-MS INTEGRATED**

**IIT-MADRAS**

**BS DATA SCIENCE AND APPLICATION**

*Interested in exploring micro robotics , quantum optics in conjunction with biophysics and neurophysics, with a focus on understanding light-matter interactions and their applications in biological and neural systems, and keen to study battery materials and energy storage as well.*

# SUMMER INTERNSHIP DESCRIPTION (2025):

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## Micro robotics Tool Development :

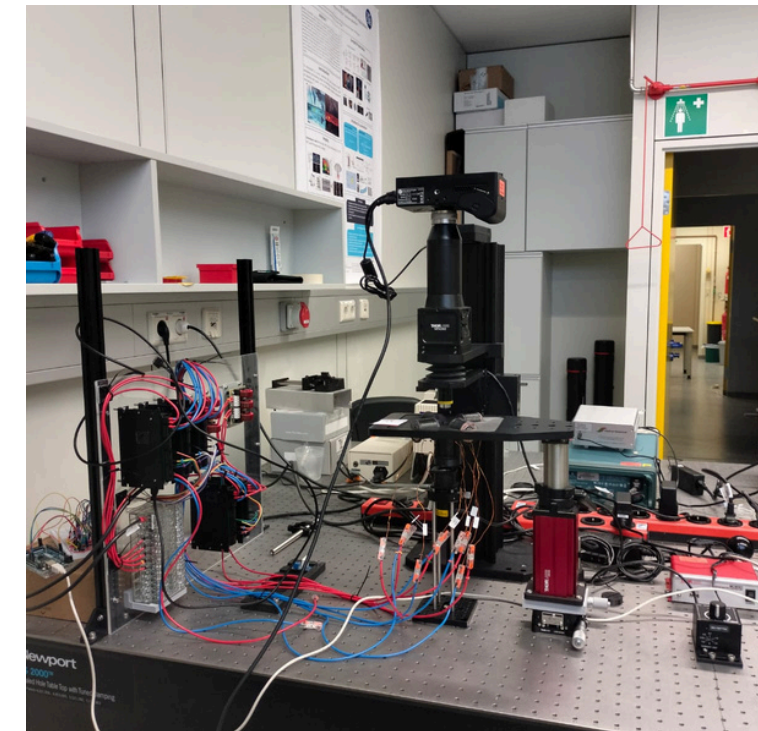
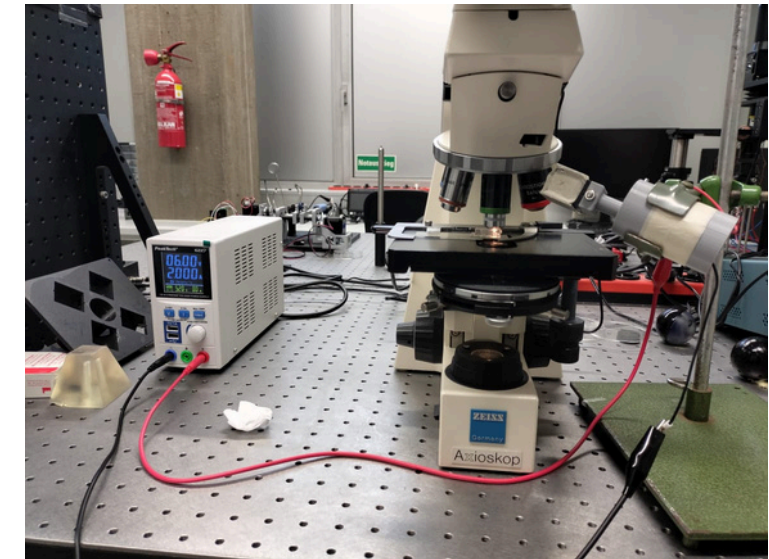
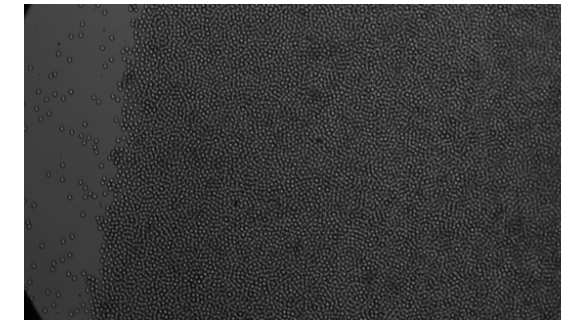
- Conducted experimental research on microrobot manipulation inside PDMS microfluidic channels bonded to glass slides.
- Utilized a five-coil electromagnetic setup alongside a permanent magnet setup for microrobot actuation and precise control of magnetic fields.
- Fabricated PDMS microchannels using Sylgard-184 and optimized bonding methods for stable microfluidic devices.
- Injected and analyzed the movement of silica and nickel microparticles (8  $\mu\text{m}$  to 50  $\mu\text{m}$ ) inside microchannels of varying sizes (100  $\mu\text{m}$  to 220  $\mu\text{m}$ ) under variable magnetic fields.
- Learned and participated in the fabrication process of hollow silica microrobots using techniques like centrifugation and sonication.

## Technical Skills Acquired :

- Microfluidic device fabrication and optimization (PDMS channel preparation, curing, bonding, and mold making).
- Operation of electromagnetic coil setup and permanent magnet setup for microrobot manipulation.
- Fabrication of hollow silica microrobots via advanced laboratory techniques (centrifugation, sonication).
- Application and control of dynamic and static magnetic fields for precise microrobot actuation.
- Experimental techniques for injection and tracking of micro/nano particles inside microchannels.
- Proficiency with microscopy, syringe pumps, and diamond cutters for microfabrication.
- Data analysis and experimental control using MATLAB programming for coil operation and system evaluation.

## Application in Microrobotics:

- Demonstrated controlled locomotion and injection of microrobots using both electromagnetic coils and permanent magnets.
- Investigated particle size and channel dimension relationships affecting microrobot mobility and adhesion.
- Developed insights into optimal magnetic field strengths (6V to 21V) to maximize microrobot movement and minimize sticking.
- Innovated microfluidic bonding techniques optimizing channel integrity and particle flow during experiments.
- Successfully fabricated hollow silica microrobots, enhancing understanding of microrobot design for biomedical and engineering applications.



## SUMMER INTERNSHIP FINAL PRESENTATION LINK (2025):

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- Here is the link of my final presentation : [Vidhi PPTS.pptx](#)

# SUMMER INTERNSHIP DESCRIPTION (ONGOING):

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## Electromagnetic Field Theory & EEG–Consciousness Analysis:

- Conducting remote research on the physical basis of consciousness under Professor Johnjoe McFadden (School of Veterinary Medicine, University of Surrey), focusing on electromagnetic field theories applied to EEG data.
- Identified and characterized EEG datasets representing normal brain activity (resting state, sleep stages) and abnormal states (seizure patterns, pathological abnormalities); clarified criteria used to define "normal" vs "abnormal" EEG.
- Used Python-based code to analyse EEG signals within the framework of conventional Maxwell's equations, reconstructing electromagnetic field behaviour from recorded neural activity.
- Implemented and applied modified (extended, complex-valued) Maxwell-type formulations to the same EEG datasets to study how extended electromagnetic field models capture neural dynamics.
- Compared field properties obtained from conventional and extended formulations (coherence, phase relationships, energy distribution) to assess their ability to represent features relevant to consciousness models.
- Regularly reviewed and synthesized advanced theoretical literature on electromagnetic field theories of consciousness, extended Maxwell's equations, and temporal electromagnetic field (tEMF) models linking field symmetries to subjective time and memory.

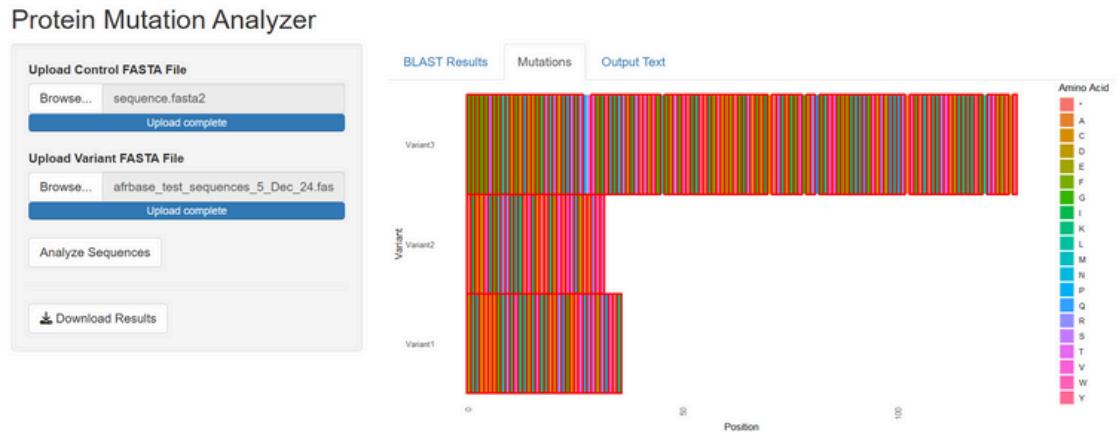
## Technical Skills Acquired :

- EEG dataset curation and characterization (normal vs abnormal; frequency-domain and time-domain features).
- EEG signal processing (filtering, spectral analysis, basic artefact handling) for use in electromagnetic modelling.
- Python-based computational modelling for applying conventional Maxwell's equations and extended complex-valued field formulations to EEG data.
- Electromagnetic field analysis (field reconstruction, coherence, phase and energy distribution comparisons) in a neurophysiological context.
- Advanced scientific literature review and theoretical synthesis on electromagnetic theories of consciousness and field-based brain models.
- Remote scientific collaboration and independent research planning in an interdisciplinary physics–neuroscience project.

# DRY LAB EXPERIENCE – BIOINFORMATICS AND QUANTUM OPTICS

## Bioinformatics Tool Development

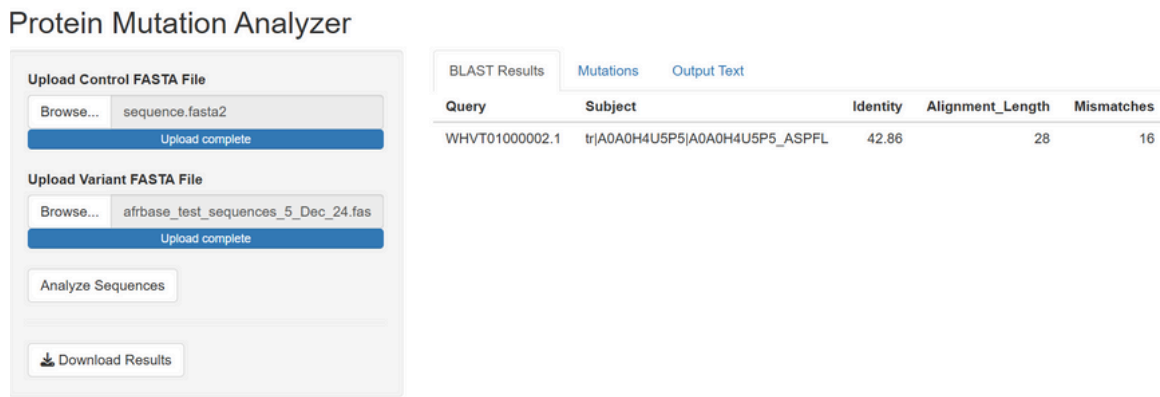
- Project: Developed a bioinformatics tool using R and Shiny for mutation analysis and drug resistance prediction.
- Key Contributions:
  - Automated sequence translation from nucleotide to protein.
  - Implemented BLAST algorithm for sequence comparison.
  - Designed mutation visualization tools for drug resistance analysis.
  - Created an interactive Shiny application for research and clinical use.
- Outcome: Delivered a user-friendly tool for mutation analysis, aiding drug development and clinical decision-making.



Final mutation visulization result

## Quantum Optics – Optical Bloch Equations

- Project: Studied light-matter interactions in two-level quantum systems during an internship at IIT Roorkee.
- Key Contributions:
  - Derived and analyzed Optical Bloch Equations (OBEs) using semi-classical approaches.
  - Simulated quantum systems using computational methods (e.g., Runge-Kutta).
  - Investigated applications in laser spectroscopy, atomic clocks, and quantum information processing.
- Outcome: Prepared a comprehensive report and presented findings on OBEs and their applications in quantum optics.



Final mutation BLAST result

# BIOPHYSICS RESEARCH – PLASMONIC NANOANTENNAS AND METAMATERIALS

## Project Overview

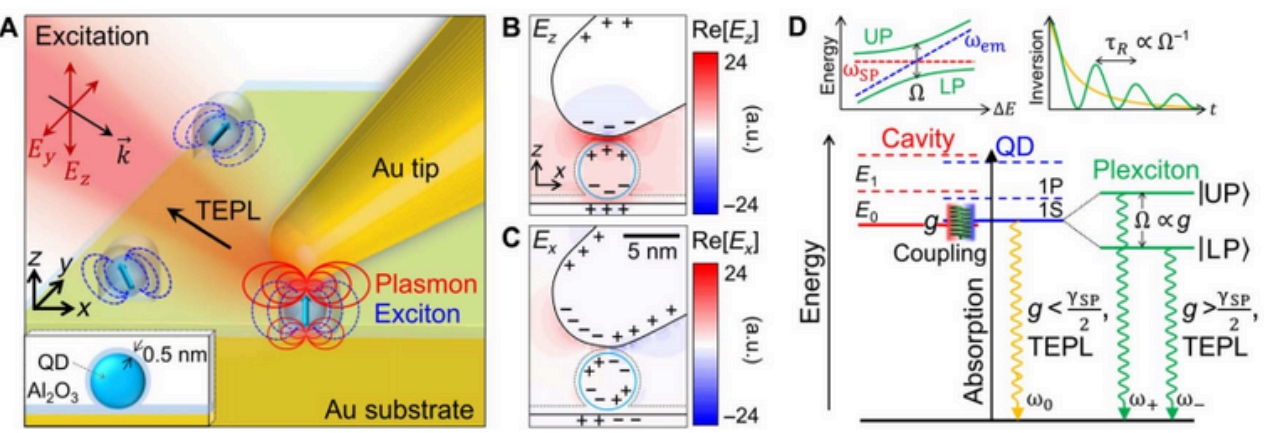
- Title: Plasmonic Nanoantennas and Metamaterials for Biochemical Analysis
- Objective: Conducted an in-depth review of advanced research papers on plasmonic nanoantennas, metamaterials, and their applications in bioinformatics and biochemical analysis.
- Description:
  - Explored the integration of nanotechnology and bioinformatics for enhanced molecular diagnostics.
  - Focused on surface-enhanced infrared absorption spectroscopy (SEIRA) and its applications in real-time biomolecular interaction studies.

## Key Contributions

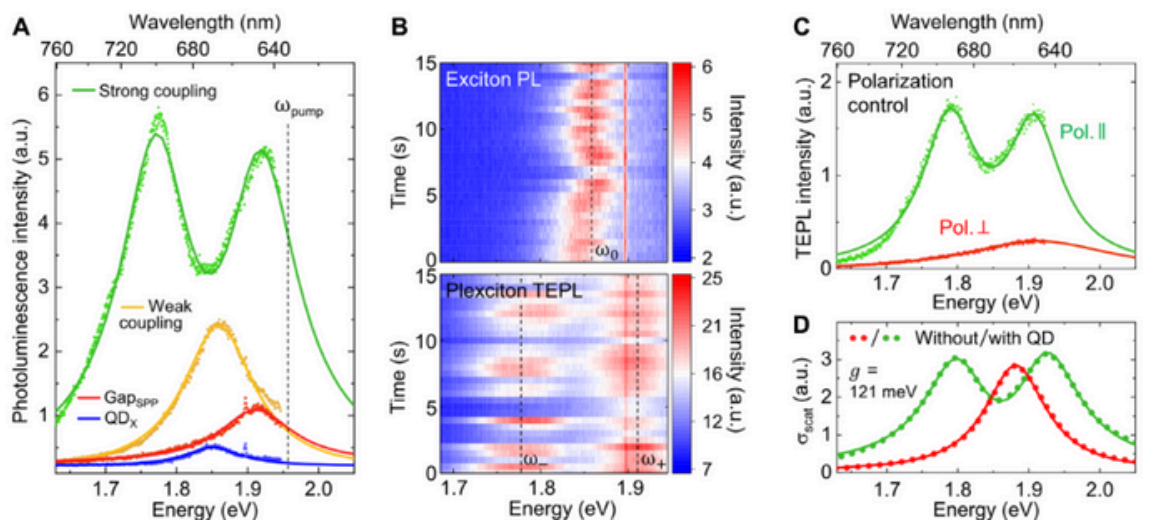
- Literature Review: Analyzed advanced research papers on plasmonic nanoantennas, Fano resonance, and metamaterial design.
- Technical Insights:
  - Investigated methodologies for integrating plasmonic nanoantennas with biochemical systems to enhance molecular detection sensitivity..
- Applications:
  - Studied the potential of SEIRA and metamaterials for protein structure analysis and biomolecular interaction studies.
  - Developed insights into how advanced spectroscopy techniques can complement computational biology.

## Skills Developed

- Scientific Literature Analysis: Proficient in reviewing and synthesizing complex research papers in nanotechnology and spectroscopy.
- Interdisciplinary Knowledge: Bridged concepts from physics, nanotechnology, and bioinformatics to explore innovative solutions for biomolecular analysis.
- Technical Communication: Presented findings effectively to a multidisciplinary audience.



TESC spectroscopy and energy diagram for the plasmon and exciton in the weak and strong coupling regime.



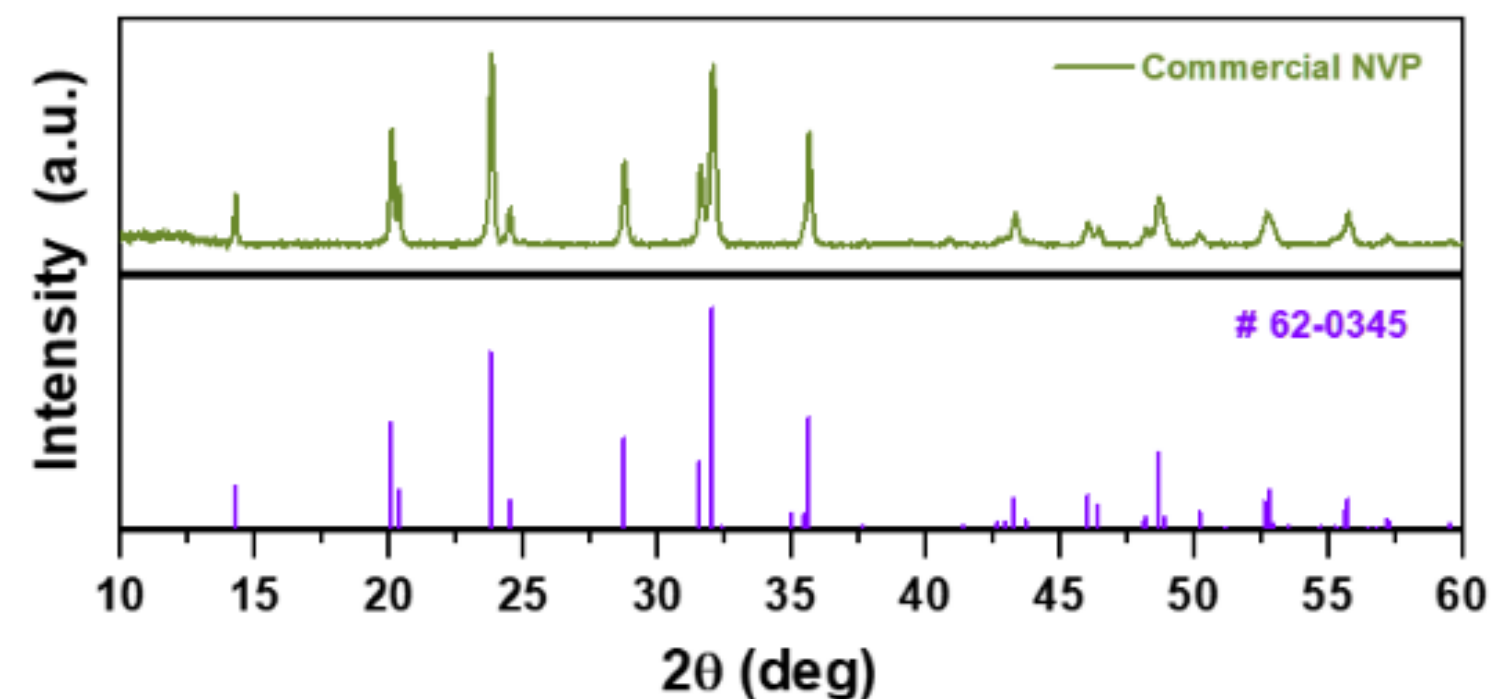
Tip-enhanced plexciton PL at room temperature in the strong coupling regime.

References:Kyoung-Duck Park et al.Tip-enhanced strong coupling spectroscopy, imaging, and control of a single quantum emitter.Sci. Adv.5,eaav5931(2019).DOI:[10.1126/sciadv.aav5931](https://doi.org/10.1126/sciadv.aav5931)

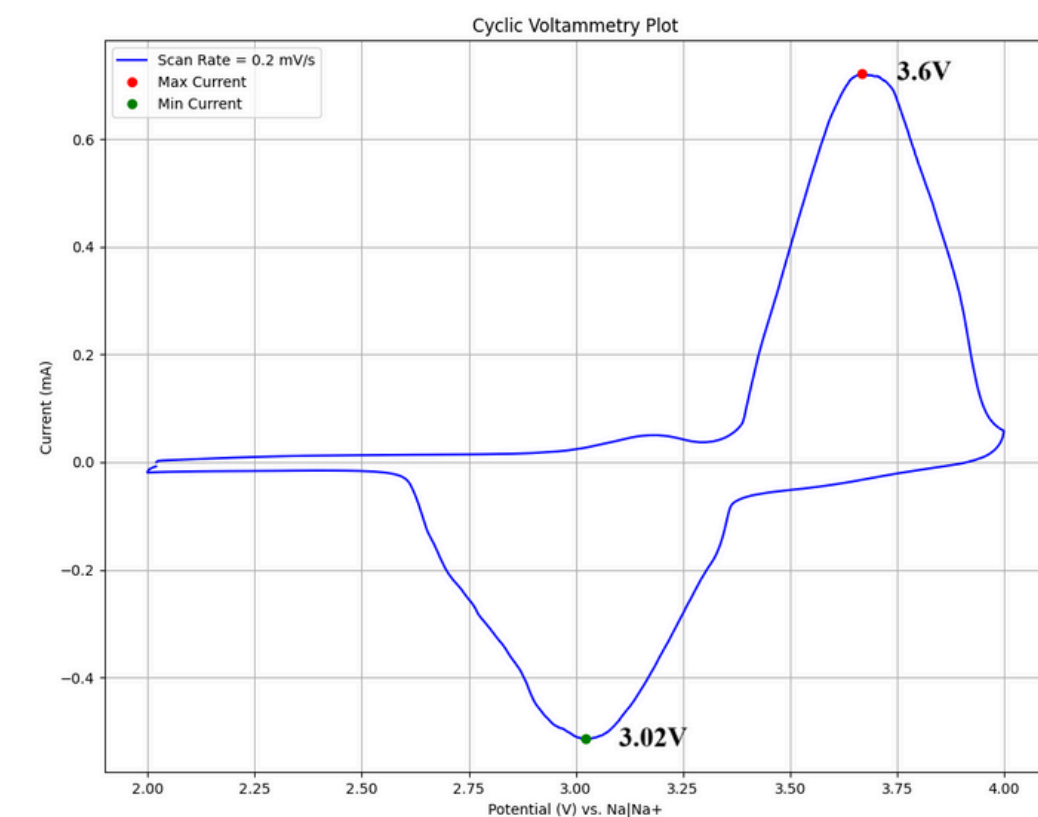
# PHYSICS RESEARCH – SODIUM VANADIUM PHOSPHATE AS CATHODE FOR SODIUM-ION BATTERIES

Research Project, IISER Thiruvananthapuram (Supervisor: Prof. M. M. Shaijumon)

- Objective: Synthesized and characterized sodium vanadium phosphate (NVP) as a cathode for sodium-ion batteries to address the limitations of lithium-ion technology.
- Key Contributions:
  - Prepared NVP cathode materials and assembled CR2032 coin cells under controlled conditions.
  - Performed XRD and SEM analyses to confirm phase purity and morphology of NVP.
  - Conducted electrochemical testing (CV, GCD, EIS) to evaluate redox activity, capacity retention, and cycling stability.
  - Demonstrated a reversible  $V^{3+}/V^{4+}$  redox reaction, with an initial capacity of 102.96 mAh/g at 0.1C and stable voltage operation at ~3.4 V.
  - Analyzed rate capability and efficiency, highlighting NVP's suitability for sustainable, large-scale energy storage.
- Skills Developed: Wet lab synthesis, battery assembly, material characterization (XRD, SEM), electrochemical analysis, data interpretation, and scientific reporting



Material Characterization- XRD



# BIOLOGY RESEARCH – DROSOPHILA POPULATION STUDY

## Project Overview

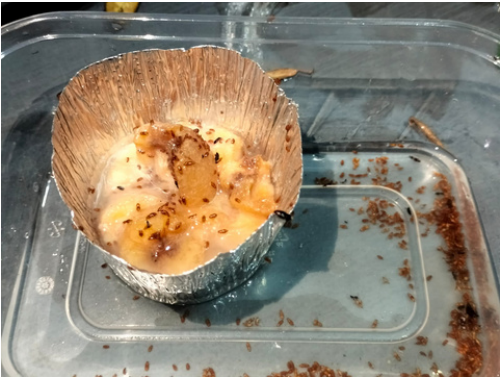
- Title: Quantifying Drosophila Populations in Disturbed/Undisturbed Environments
- Objective: Investigated the impact of environmental factors (disturbance and hygiene) on Drosophila populations.
- Description:
  - Designed a controlled field experiment with replicated trapping in disturbed/undisturbed and hygienic/unhygienic locations.
  - Analyzed population variations to understand ecological connections.

## Key Contributions

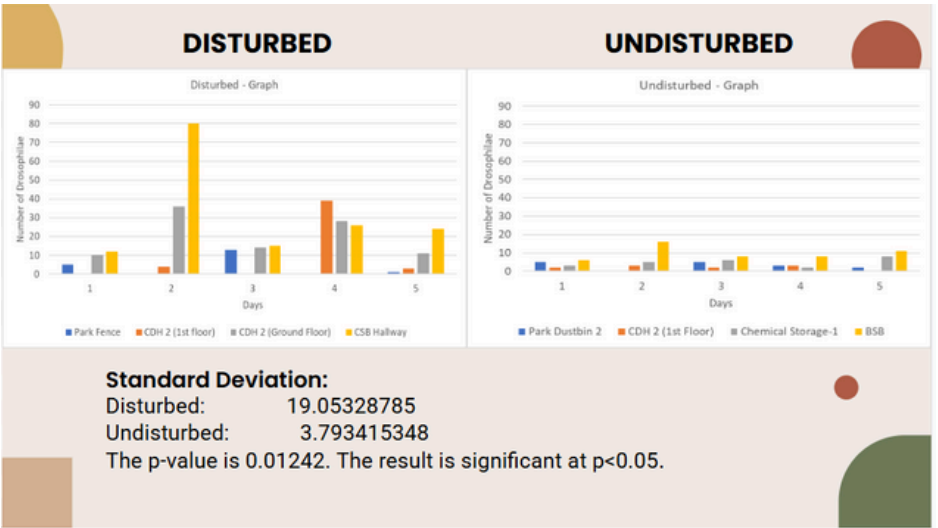
- Experimental Design: Created a controlled setup to study environmental impacts on Drosophila populations.
- Data Collection: Conducted field experiments and collected population data.
- Data Analysis: Analyzed population variations and identified trends related to disturbance and hygiene.

## Skills Developed

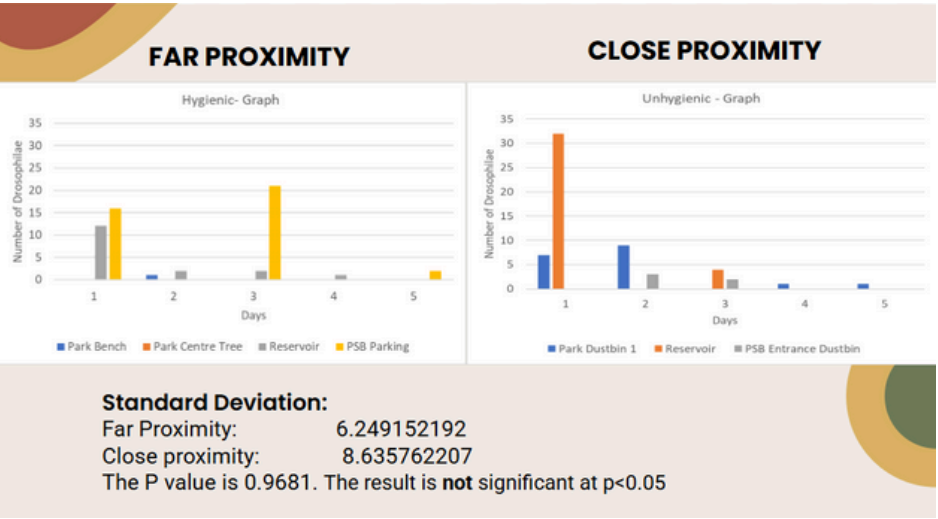
- Field Research: Gained hands-on experience in designing and conducting ecological experiments.
- Data Analysis: Analyzed ecological data to draw meaningful conclusions.
- Scientific Communication: Presented findings on population dynamics and ecological connections.



(Photograph of the experimental setup with bait used for trapping Drosophila, illustrating the controlled field experiment design.)



(Comparison of Drosophila populations in disturbed and undisturbed environments, showing significant variations influenced by environmental conditions ( $p < 0.05$ ).)



(Analysis of population density based on proximity to hygienic and unhygienic sites, highlighting the impact of spatial factors on Drosophila distribution ( $p < 0.05$ ).)

# WET LAB EXPERIENCE – BIOLOGY LABORATORY COURSES

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## Courses Completed:

- BIO112 (Biology Laboratory I):
  - Hypothesis testing, sampling, microscopy, DCPIP method for photosynthesis, microbial isolation, Gram staining, plaque assay.
- BIO122 (Biology Laboratory II):
  - Preparation of biological solutions, quantification of biomolecules (proteins, lipids, carbohydrates, DNA), genomic DNA isolation, PCR, enzyme assays.
- BIO212 (Biology Laboratory III):
  - Mutation frequencies, plasmid DNA isolation, SDS-PAGE, mitosis, and meiosis analysis.

## Key Wet Lab Skills:

- Microscopy: Plant and animal cell observation, tissue structure analysis.
- Molecular Techniques: DNA isolation (genomic and plasmid), PCR, SDS-PAGE.
- Biochemical Assays: Enzyme assays, quantification of biomolecules, DCPIP method for photosynthesis.
- Microbiology: Microbial isolation, Gram staining, plaque assays.

# PHYSICS RESEARCH – CASIMIR EFFECT

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## Project Overview

- Title: Casimir Effect and Water Wave Analog
- Objective: Explored the Casimir Effect, a quantum mechanics phenomenon, using a water wave analog.
- Description:
  - The Casimir Effect predicts an attractive force between two closely spaced conducting plates due to quantum vacuum fluctuations.
  - Designed and conducted a water wave experiment to demonstrate the effect of confined wave patterns on force generation.

## Key Contributions

- Experimental Design: Developed a water wave setup to simulate the Casimir Effect.
- Data Analysis: Analyzed wave patterns and their impact on force generation.
- Presentation: Presented findings at IISER Thiruvananthapuram’s selfest festival.

## Skills Developed

- Experimental Physics: Gained hands-on experience in designing and conducting physics experiments.
- Data Interpretation: Analyzed wave patterns to explain quantum phenomena.
- Scientific Communication: Effectively communicate complex concepts to a non-specialist audience.



Top view of the experimental setup showing the confined water wave patterns created between closely spaced plates, simulating the Casimir Effect analog.



Closer view highlighting the interference patterns of water waves, demonstrating the impact of confinement on wave behavior and force generation.



Side view of the experimental apparatus, illustrating the alignment and spacing of plates used to replicate the quantum vacuum fluctuation analogy.

# SKILL SETS

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## Technical Skills

- Programming: Proficient in C, C++, Python, and R.
- Data Analysis: Experienced in computational modeling, sequence analysis, and mutation visualization.
- Quantum Optics: Skilled in Optical Bloch Equations, quantum system simulations, and light-matter interactions.
- Bioinformatics: Developed tools for mutation analysis and drug resistance prediction using R Shiny.

## Wet Lab Skills

- Molecular Techniques: DNA isolation (genomic and plasmid), PCR, SDS-PAGE, and enzyme assays.
- Microscopy: Plant and animal cell observation, tissue structure analysis.
- Microbiology: Microbial isolation, Gram staining, and plaque assays.

## Research and Analytical Skills

- Experimental Design: Designed and conducted experiments in physics (Casimir Effect) and biology (Drosophila population study).
- Data Interpretation: Analyzed ecological, quantum, and bioinformatics data to draw meaningful conclusions.
- Scientific Communication: Presented findings at academic events and prepared comprehensive reports.

## Soft Skills

- Teamwork: Collaborated with peers and mentors on interdisciplinary projects.
- Problem-solving: Applied analytical and computational skills to solve complex scientific problems.
- Time Management: Balanced coursework, research, and extracurricular activities effectively.

- Thank you for reviewing my application. I look forward to hearing from you soon.

- Contact Information:

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